in the desktop graphics arena from AMD who beat them to market with DX 11 parts. NVIDIA fought back with the GTX 460 and later, the 5xx series, at around the same time Tegra 2 happened. This SoC follows the nomenclature of Txx (xx being a number) for tablets, and APxx for phones.

NVIDIA's strength lies in the GPU domain, and while the Cortex-A9 is the architecture to beat at the moment, the GPU in the Tegra 250 is around 50-60 per cent faster than the Snapdragon Adreno 205. It is also a lot more powerful than the PowerVR SGX 540 GPU, which Samsung and Apple use.

The LG Optimus 2x running NVIDIA Tegra 2 hardware

At CES 2011, NVIDIA announced an SoC codenamed Kal-El (Superman's birth name). Kal-El will be built on TSMC's proven 40nm fabrication process. A 12-core GPU is also on the cards, which NVIDIA claims will be five times more powerful than the GeForce ULP on the Tegra 2. This SoC will be capable of Blu-Ray playback as well as native 1080p HD content, and resolution support is up to 2560 x 1600 pixels – overkill perhaps, but at no loss to battery life it will be fantastic.

A month later, at the MWC another SoC called "Wayne" was announced. This time on a 28nm fab process. The GPU is claimed to be 12 times more powerful than Tegra 2. Two other SoCs were announced, but other than their obvious superhero-inspired names (Logan and Stark), not much else is known, except that NVIDIA claims 75 times better performance from Stark, when compared to Tegra 2.

Samsung (and Apple): Hummingbird, A4, A5 and Orion

Samsung manufactures Apple's SoC, and the A4 and A5 chips are Samsung made. These are loosely based on Samsung's own SoC called Hummingbird. Samsung made these chips with the help of a small company called Intrinsity. The Hummingbird includes ARM's own SIMD engine, called ARM NEON.

This SoC contains the PowerVR SGX535, which is a powerful gaming platform, as seen in the iPhone 4. While the Adreno 205 is rumoured to be slightly more powerful, Apple definitely has a vast ecosystem of optimised games. The Hummingbird/A4 SoC does not contain RAM to keep its cost down. Additionally there is no baseband unit – Qualcomm integrates this component.

At the MWC Samsung launched their Galaxy S2, a smartphone based on an SoC called Exynos, which was earlier called Orion. ARM's NEON is once again part of the package, and this time GPS is integrated. The Mali 400 GPU in this SoC is an ARM design, and a product of ARM's fledgling graphics division, although it is an older GPU.


The much awaited Samsung Galaxy S2, featuring the new Samsung Exynos SoC, 1 GB of RAM, and ARM's Mali 400 GPU

Apple also recently launched the iPad 2, based on the A5 SoC, which is basically a slightly modified Exynos.

Samsung is also working with NVIDIA, obviously to strengthen the graphics portion of their SoC. These two might also team up for smartphones, a move Samsung is more keen on. Exynos with GeForce graphics (if it happens) should be a killer performance package.

Texas Instruments: OMAP

TI calls their SoC family OMAP, that stands for Open Multimedia Application Platform. There are three main families of SoC for smartphones - OMAP 3, OMAP

	2011				2012		
U8500	Snapdragon (3rd Gen)	A9600/A9500/A9450	Exynos (formerly Orion)	Kal-El	4th Gen Snapdragon	OMAP 5	Wayne
ST-Ericsson	Qualcomm	ST-Ericsson	Samsung	NVIDIA	Qualcomm	Texas Instruments	NVIDIA
ARMv7a	ARMv7a	ARMv7a	ARMv7a	ARMv7a	ARMv7a	ARMv7a	ARMv7a
Cortex-A9	Scorpion	-	Exynos	Cortex-A9	Krait	-	-
Cortex-A9	ARM Cortex-A8	ARM Cortex-A9 / A15	ARM Cortex-A9	Cortex-A9	Unknown	ARM Cortex-A15 / M4	ARM Cortex-A15
2	1, 2	2, 4	2	2, 4	1, 2, 4	4	4
1.2 GHz	1.2 GHz - 1.5 GHz	1.2 GHz - 2.5 GHz	1 GHz and above	1.5 GHz and above	Up to 2.5 GHz (lower clock speeds unknown)	2 GHz	not known
ARM MALI 400	Adreno 220	ARM Mali 400, PowerVR 6 series (codenamed "Rouge")	ARM MALI 400	12 core GeForce ULP	Adreno 305, 320	PowerVR SGX544MP	GeForce ULP (details unknown)
U8500, U5500	MSM8260, MSM8660, APQ8060	A9600, A9500, A9450	Exynos 4210	not yet announced	APQ8064, MSM8960, MSM8930	OMAP 5430, 5432	-
-		-	Samsung Galaxy S2	-	-	-	-
-	HP Touchpad, ASUS Eee Pad MeMo	-	Apple iPad 2.0 (modified A5 core)	-	-	-	-
This SoC would be the dark horse, for while it has no takers currently, we could see Nokia using it to build an army of WP7 devices	There hasn't been a lot of devices on this SoC as of now, probably because cellphones are yet to fully utilise dual cores	The A9600 is Cortex-A15 based, and should be seen in action on tablets. The A9500 is a product aimed at cellphones, while the A9450 might also go the tablet way, since it features dual core A9s clocked at a hefty 1.8 GHz	We were all awaiting the successor to the Galaxy S, and the S2 was announced using this SoC. Perhaps, that has helped generate more interest in the Exynos	Kal-El could be an iPad killer if eveything works out well for NVIDIA. The problem is going to be getting it into a highly developed device with a worthy interface, and Apple doesn't have much competition as of yet	This is where Qualcomm really starts for focus on graphics and processing power - not much else is known, but with the competition heating up, Qualcomm is expected to pull something special out of the bag	OMAPs entry into the tablet segment with quad cores. The only difference between both SoC is the DDR2 controller on the OMAP5430 and the DDR3 controller on the OMAP5432	Wayne will be the first SoC NVIDIA makes on a 28nm fabrication process, which means power and space savings